Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/28/2004
PI Name:	Rabolt, John F. Ph.D.		
Project Title:	Synthesis, Characterization and Processing of Genetically Engineered Polymers for Biological and Structural Applications		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	06/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	Note: Original end date was 5/31/2007.		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Kiick, K. (MSE, University of Delaware)		
Grant/Contract No.:	NAG8-1923		
Performance Goal No.:			
Performance Goal Text:			
	The work being undertaken will involve synthesis, characterization, processing and the evaluation of structure/property relationships. Our overall strategy involves the design and production of protein polymers according to their desired NASA-relevant biological and/or structural application. The genetically engineered polymers will then be characterized in solution prior to their being spun into fibers by electrospinning methods. These studies will allow us to assess the role of self assembly (e.g., a-helix, ß-sheet) on the resulting microstructure, orientation, surface morphology, and properties of electrospun fibers. In addition, in order to optimize structure-processing-property relationships we plan to use Raman spectroscopy to characterize the electrospinning fiber as it is being formed from the liquid polymer jet. Using this technique to obtain “real-time” information on the extent of orientation and the ratio of solvent/polymer at different points within the spinning jet will allow us begin to correlate the processing protocols (voltage, distance from droplet to		

Task Description:	target, solution viscosity, etc.) with the resulting fiber structure/morphology. The described combination of polymer design, synthesis, characterization, and processing will permit the purposeful formation of bioactive fiber mats that could be used in wound healing applications while protecting against inflammation and infection. Genetic engineering of advanced materials for biological (tissue engineering, wound repair, scaffolds), electronic (fiber optics, waveguides, active devices), and structural (high tenacity, high modulus) applications will require significant advances in the synthesis, characterization, and processing of bio-inspired materials during the next decade. Coupled with the requirement for new chemical strategies for bio-inspired materials synthesis is the need for advanced processing protocols to provide the morphology and shape control required for these diverse applications. The goals of this proposed work are to both develop new biologically derived materials and to implement novel processing strategies in order to control the macroscopic shape, the polymer backbone orientation/conformation and the surface topography/morphology of the materials produced. Such control will allow optimization of the materials structure/property relationships for biological and structural applications of interest to NASA. Two specific synthetic routes that capitalize on genetically directed synthesis of a protein polymer will be pursued. Novel electrospinning methods will be combined with real-time characterization to allow manipulation of the morphology and shape of these unique protein-based materials and could potentially be used as the basis of a compact, portable wound repair and/or tissue scaffold system in the context of remote space travel where a rapid return to Earth is not possible. The protein architectures that are being synthesized in this work have been chosen for their synthetic flexibility and for their ability to be derivatized with a variety of saccharide ligands whose multivalent presentation controls many biological processes. For example, the multivalent presentation of galactose ligands controls bacterial chemotaxis and could therefore control bacterial infection. Multivalent presentation of sialic acid has applications for controlling viral infection, and presentation of the Lewis X trisaccharide has opportunities for limiting inflammation. Importantly, these underivatized helical polymers alone may serve as a scaffold for the organization of the endothelium, which suggests that the molecules could find a variety of uses in controlling wound healing, infection, and/or inflammation, as spun into fiber mats either separately or in mixtures. Such skin regeneration and wound healing applications would be important for controlling physiological responses to serious injuries, such as large area burns, sustained by NASA crew members.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Newly designed protein polymers have been synthesized by recombinant DNA methods. They have been electrospun into macro- and fibrous membranes for use as tissue engineering scaffolds. These generic constructs can be used in many diverse applications that range from wound repair to the fabrication of a cardiac patch. More than 700,000 Americans die each year from heart disease at a cost of $238B in health care related expenses. Hence the outcome of this work has the potential to have significant impact on the health of society.
Task Progress:	During the first year of the grant, two research efforts were initiated: 1) recombinant DNA synthesis of an a-helical protein polymer with functionalized side chains located at periodic distances along the backbone; and 2) electrospinning of genetically engineered spider silk synthesized by Dr. Stephen Fahnestock at DuPont. We have been using genetically directed synthetic methods to produce a number of unique protein polymers and then electrospin them into nanofibers so as to assess their mechanical integrity and biological activity. Significant progress has been made on the electrospinning of polymer nanofibers [1]. Dragline spider silk is one of the strongest materials known, yielding properties that are comparable, by weight, to high strength steel. The basis for this is due, in part, to the amino acid sequence of the silk fibroin (silk protein). It can be thought of as a block-like copolymer with hard/soft segment repeat units, forming a hard crystalline structure that is embedded in an elastic matrix. Raman spectroscopy of the fibrous membrane has revealed that, although the as-synthesized spider silk analogue adopts a ß-sheet conformation, electrospinning produces nanofibers that adopt predominantly an a-helical structure. This was very surprising because no such conformational transition has been reported in the literature. Starting with a solid film of a-helical poly(L-alanine), Frushour and Koenig [2] showed that a ß-sheet conformation could be induced by drawing, but not the reverse. Hence our hypothesis at this point is that, since spider silk is electrospun from hexafluoroisopropanol, it adopts a partially a-helical conformation in solution. Since the hydrogen bonds in an a-helix are along the helical axis, the molecular dipoles will be additive and hence the resulting dipole moment can be sizeable for six alanine residues. The large electric field applied during the electrospinning process could then cause the self-assembly of these dipolar segments with the resulting preservation of the a-helical structure. Current work to verify or disprove this hypothesis will be undertaken in year 2 of this grant. Recently, polysaccharide-derivatized PEG star polymers have been synthesized and are currently being evaluated as a strategy for including functional macromolecules into electrospun fibers. PEG-star polymers have been derivatized with the sulfated, helical polysaccharide heparin, and these molecules have been incorporated into electrospun PEO fibers. Heparin alone cannot be incorporated into electrospun PEO fibers; attachment of the polysaccharide to a PEG scaffold is required in order to incorporate the biomolecule into the fiber. Alteration of the spinning conditions results in radically different fiber morphologies; the origins of these differences are being explored, along with the potential for orientation of the helical heparin molecules. These preliminary results suggest the utility of copolymers as a means to produce novel functionalized electrospun fibers. References [1] Stephens, J. S.; Rabolt, J. F.; Fahnestock, S.R. and Chase, D. B. “Effects of Electrospinning on the Secondary Structure of a Dragline Spider Silk Analog Investigated by FT-Raman Spectroscopy” Biomacromolecules (submitted 5/04) [2] Frushour, B.G. and Koenig, J.L., Biopolymers 3, 455 (1974)
Bibliography Type:	Description: (Last Updated: 04/17/2018)

Articles in Peer-reviewed Journals	Stephens JS, Fahnestock SR, Farmer RS, Kiick KL, Chase DB, Rabolt JF. "Effects of electrospinning and solution casting protocols on the secondary structure of a genetically engineered dragline spider silk analogue investigated via Fourier transform Raman spectroscopy." Biomacromolecules. 2005 May-Jun;6(3):1405-13. PubMed PMID: 15877359 ; https://doi.org/10.1021/bm049296h , Jun-2005